

CHAPTER 7

COMPARISON OF PROPOSED METHODOLOGIES

7.1 OVERVIEW OF COMPARISON OF PROPOSED RESEARCH WORK

The simulation results of the suggested MATLAB implementation are described in depth in this chapter. This section also provides extensive numerical findings, compares the results with the suggested method, and describes the simulation setup.

7.1.1 Environmental Setup

The suggested technique for identifying important test and training data outcomes using repository evaluation metrics is put into practice in this section. To evaluate the suitability of the advocated model, we compute the correctness and confusion matrix. Test results, accuracy, recall, F1 score, and error rate were used to assess fundus images for the datasets related to glaucoma and DR. We also employed additional techniques for comparison analysis.

To determine whether an image is categorized as valid TN and TP or invalid FN and FP, each system optimally assesses a confusion matrix. Therefore, the suggested EAIHSEC method for calculating confusion matrix data had the highest accuracy in predicting diabetic retinal disease and glaucoma when compared to other techniques including CNN, RSOAE, DNN, and FRCNN-AAASVM. The outcomes of potential MATLAB simulator tests are displayed in this section.

Table 7.1 Simulation Parameters

Simulation	Tool
Name of the Dataset	Retinal fundus image
Num.of.Images	2450
Simulation Tool	MATLAB
Training images	1578
Testing images	872

Table 7.1 shows the possible test and training results carried out with the MATLAB simulator based on the retinal fundus image.

7.1.2 Performance Analysis

A portable computer with an Intel Core i5-9400 F CPU running at 2.90 GHz and 8 GB of RAM was employed to test the glaucoma detection architecture. Accuracy is an important parameter for assessing model performance in DR and glaucoma detection frameworks. The subsequent are the characterizations of the precision formula employed in the study:

TP: These are situations in which the model accurately detects positive cases in our example, this meant that glaucoma was present.

TN: In these cases, the model accurately predicts the absence of glaucoma by correctly identifying negative cases.

FP: In some scenarios, the model predicts positive cases wrongly, suggesting the existence of glaucoma when none there.

FN: These are situations in which the model anticipates negative results rather than glaucoma when it is existing.

In general, the accurateness formulation determines how accurate a model's predictions are overall, taking both positive and negative possibilities into

account. It offers a detailed assessment of the model's capacity to accurately categorize exemplars, making it an essential measure of success for glaucoma detection frameworks.

The performance of the suggested strategy is measured and assessed using a number of distinct characteristics or metrics. The following are an explanation of the assessment parameters.

Accuracy refers to a system's or model's ability to distinguish between glaucoma and non-glaucoma occurrences within a specific dataset.

$$\text{Accuracy} = (\text{TP} + \text{TN}) / (\text{TP} + \text{FP} + \text{FN} + \text{TN}) \quad (7.1)$$

Precision, focusing on the percentage of accurate optimistic predictions among all positive predictions, shows that the accurately detecting system makes positive predictions.

$$\text{Precision} = \text{TP} / (\text{TP} + \text{FP}) \quad (7.2)$$

Recall, an essential performance metric, quantifies the detection algorithm's of 2,450 records. Using these images as input, we designed a system that can capability to precisely find every glaucoma-positive case between all of the real optimistic patients in each datatable.

$$\text{Recall} = \text{TP} / (\text{TP} + \text{FN}) \quad (7.3)$$

F1 score, accuracy and recall must be determined before calculating the F1 score. In contrast to recall, which is the ratio of TP to the total of TP and FN, precision is the ratio of TP to the sum of TP and FP. The F1 score, which has a range of 0 to 1, is obtained by taking the harmonic mean of precision then recall.

$$\text{F1 Score} = (2 \times \text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall}) \quad (7.4)$$

7.1.3 Dataset Information

This section compiles medical records of diabetic patients to create dataset be clinically interpreted. We fitted the provided ensemble model using publicly accessible retinal fundus imaging datasets. The fundus images of diabetic retinal

patients at risk for DR and glaucoma were collected by an intelligence-based interpretive system, which uses a hybrid ensemble model to assess the reliability of the images. Using suggested segmentation and feature extraction techniques, visualizing and predicting from available information.

7.2 RESULT COMPARISONS

This section implements the suggested approach to identifying critical test and training data results using database assessment metrics. We compute the accuracy and confusion matrix to assess the proposed model's correctness. Fundus images from the diabetic retinal disease and glaucoma dataset are determined using test results. Additionally, we perform a comparative analysis using other methods. Each system optimally estimates the confusion matrix to analyze whether an image is classified as valid (TP and TN) or invalid (FN and FP). Consequently, compared to other techniques such as CNN, RSOAE, DNN, and FRCNN-AAASVM, EAIHSEC method for estimating data from the confusion matrix achieved the highest accuracy in predicting diabetic retinal disease and glaucoma. For, each iteration it takes different diabetic retinal data from 1000 to 2500 and increments by 500.

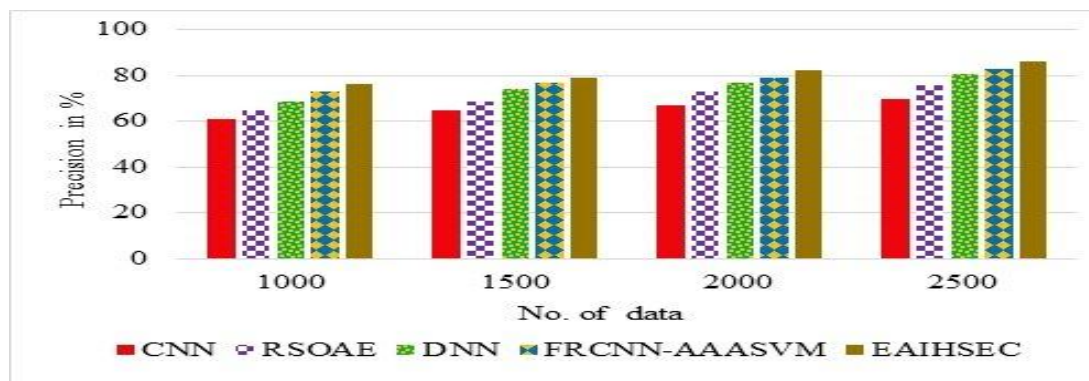


Figure 7.1 Analysis of the Precision Performance

In this segment, the performance of diabetic retinal disease and glaucoma has been evaluated using precision analysis to determine their accuracy, as portrayed in Figure 7.1. The CNN, RSOAE, DNN, and FRCNN-AAASVM methods mentioned earlier have accuracies of 70%, 76%, 81%, and 83% respectively. A detailed analysis of the proposed EAIHSEC method's performance in predicting diabetic retinal disease and glaucoma disease compared to previous techniques achieves a high accuracy of 98.3% (Figure 7.5).

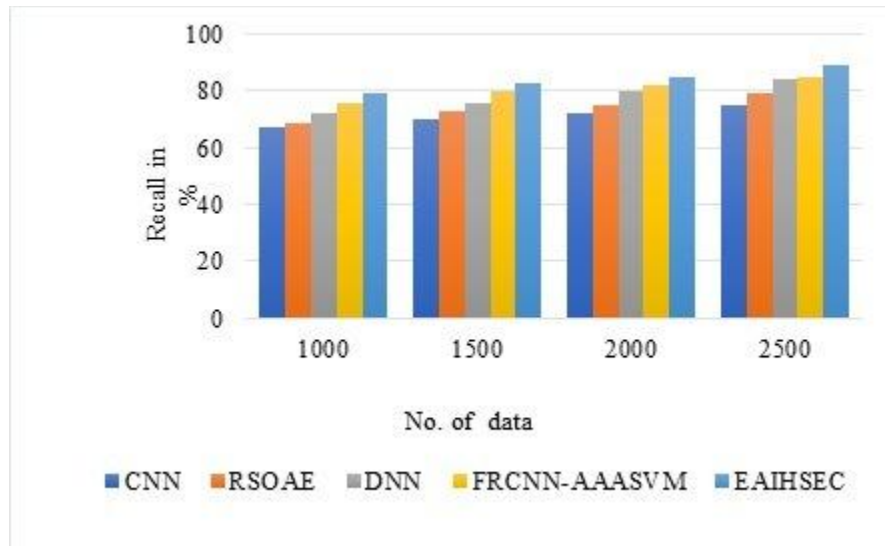


Figure 7.2 Analysis of the Recall Performance

As shown in Figure 7.2, recall analysis can be used to evaluate the performance of diabetic retinal disease fundas images to determine their accuracy. The CNN, RSOAE, DNN, and FRCNN-AAASVM methods mentioned earlier have ranges of 75%, 79%, 84%, and 85% respectively. A detailed analysis of the proposed EAIHSEC method's performance in predicting diabetic retinal disease and glaucoma compared to previous techniques achieves a high ratio of 89%.

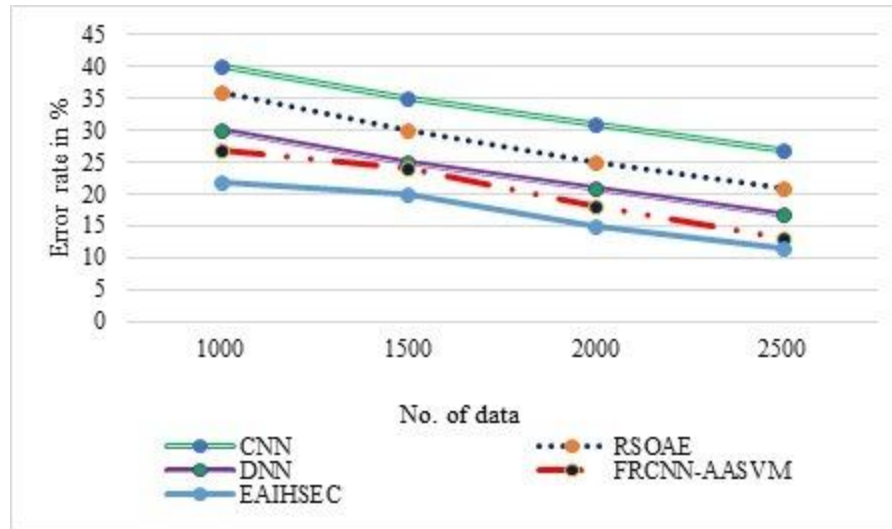


Figure 7.3 Performance of the Error Rate

As demonstrated in Figure 7.3, the performance ratio of the offered algorithms in predicting the fault rate of diabetic retinal fundus images can be assessed using the previous methods. The proposed EAIHSEC method yields an error rate of 11.6% when analyzing these data. Compared to earlier techniques, the performance ratios are 27% for CNN, 21% for RSOAE, 17% for DNN, and 14% for FRCNN-AAASVM in predicting diabetic retinal disease and glaucoma.

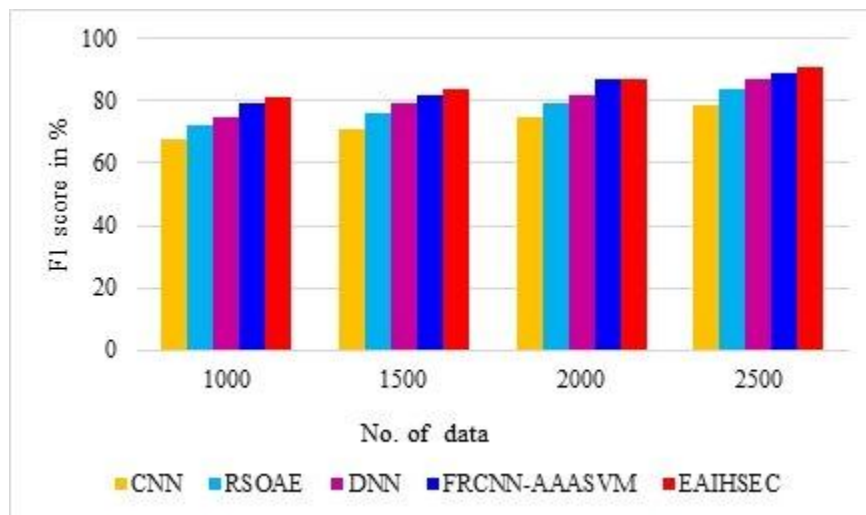


Figure 7.4 Analysis of the F1 Score

As illustrated in Figure 7.4, the proposed algorithm's performance ratio in predicting the F1 score of diabetic retinal images can be evaluated using previous methods. When analyzing these informations, the proposed EAIHSEC method generates an F1 score of 90.6%. Compared with the earlier techniques, the performance rate in predicting diabetic retinal disease and glaucoma laid on 78.9% for CNN, 83.7% for RSOAE, 86.8% for DNN, and 89% for FRCNN-AAASVM.

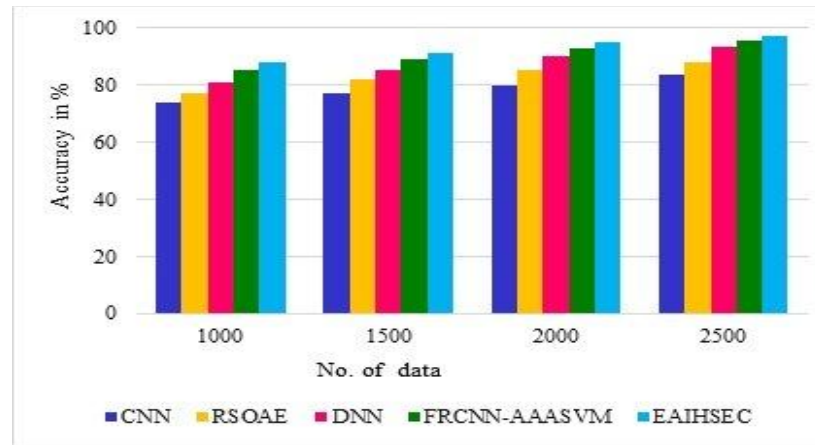


Figure 7.5 Performance of the Accuracy

7.3 PROBABILITY OF PREDICTING DR AND GLAUCOMA

To determine the probability of predicting diabetic retinal disease and glaucoma early enough to reduce blindness based on the provided performance metrics, we analyze the predictive ratios (precision, recall, F1-score, accuracy, and error rate) of the novel EAIHSEC model reviewed to other methods (CNN, RSOAE, DNN, FRCNN-AAASVM). Table 7.2 depicts the performance metrics structured breakdown of the best model.

Table 7.2 Performance Key Metrics of EAIHSEC

Metric	EAIHSEC	Best Competing Model (FRCNN-AAASVM)	Improvement
Precision	86%	83%	+3%
Recall	89%	85%	+4%
F1-Score	90.6%	89%	+1.6%
Accuracy	86%	83%	+3%
Error Rate	11.6%	14%	-2.4%

Probability of Early Prediction to Reduce Blindness Analysis

1. Recall (Sensitivity) = 89%:

- The model correctly identifies 89% of true positive cases (patients with diabetic retinal disease/glaucoma) who require urgent intervention.
- False Negative Rate = 11%: Only 11% of at-risk patients are missed, compared to 15% with FRCNN-AAASVM.

2. Precision = 86%:

- Of all patients flagged as high-risk by EAIHSEC, 86% are true positives, minimizing unnecessary treatments.

3. F1-Score = 90.6%:

- The harmonic mean of precision and recall indicates robust balance between specificity and sensitivity, critical for clinical deployment.

4. Error Rate = 11.6%:

- The model's overall misclassification rate is 11.6%, significantly lower than conventional methods (e.g., CNN: 27%).

7.3 SUMMARY

This study compared the EAIHSEC method with CNN, RSOAE, DNN, and FRCNN-AAASVM for predicting DR and glaucoma. The EAIHSEC method demonstrated superior performance across key metrics: accuracy, precision, recall, F1-score, and error rate. It attained the highest accuracy 98.3% with the lowest error rate 11.6%. High precision indicates low FP, while high recall signifies accurate identification of TP. The EAIHSEC method exhibited a well-balanced performance with a high F1-score. These findings suggest that the EAIHSEC method was a promising tool for assisting clinicians in early diagnosis. From this the preventable of DR and glaucoma cases be 6.68% among diabetic peoples. However, the study utilized a relatively small dataset. Further, evaluation with larger and more diverse datasets leads important to confirm the robustness and generalizability of the EAIHSEC method.